Matgeo Presentation - Problem 5.2.56

EE25BTECH11054 - S.Harsha Vardhan

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Problem Statement

Solve the following system of linear equations.

$$\frac{5}{x-1} + \frac{1}{y-2} = 2 \quad \frac{6}{x-1} - \frac{3}{y-2} = 1$$

Solution: Substitution

Let

$$u = \frac{1}{x-1}, \quad v = \frac{1}{y-2}. \tag{0.1}$$

The given system becomes a linear system in terms of u and v:

$$5u + v = 2 \tag{0.2}$$

$$6u - 3v = 1 \tag{0.3}$$

Solution: Matrix Form

In matrix form, this is $\mathbf{A}\mathbf{x} = \mathbf{b}$, where:

$$\begin{pmatrix} 5 & 1 \\ 6 & -3 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}. \tag{0.4}$$

Let

$$\mathbf{A} = \begin{pmatrix} 5 & 1 \\ 6 & -3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}. \tag{0.5}$$

We can solve this using Gauss-Jordan elimination on the augmented matrix $\left(\mathbf{A}|\mathbf{b}\right)$.

Solution: Gauss-Jordan Elimination

$$\left(\begin{array}{cc|c} 5 & 1 & 2 \\ 6 & -3 & 1 \end{array}\right) \xrightarrow{R_1 \to \frac{1}{5}R_1} \left(\begin{array}{cc|c} 1 & \frac{1}{5} & \frac{2}{5} \\ 6 & -3 & 1 \end{array}\right) \xrightarrow{R_2 \to R_2 - 6R_1} \left(\begin{array}{cc|c} 1 & \frac{1}{5} & \frac{2}{5} \\ 0 & -\frac{21}{5} & -\frac{7}{5} \end{array}\right) \tag{0.6}$$

Solution: Gauss-Jordan Elimination

$$\left(\begin{array}{cc|c} 1 & \frac{1}{5} & \frac{2}{5} \\ 0 & -\frac{21}{5} & -\frac{7}{5} \end{array}\right) \xrightarrow{R_2 \to -\frac{5}{21} R_2} \left(\begin{array}{cc|c} 1 & \frac{1}{5} & \frac{2}{5} \\ 0 & 1 & \frac{1}{3} \end{array}\right) \xrightarrow{R_1 \to R_1 - \frac{1}{5} R_2} \left(\begin{array}{cc|c} 1 & 0 & \frac{1}{3} \\ 0 & 1 & \frac{1}{3} \end{array}\right) \tag{0.7}$$

Solution: Back-Substitution

From the reduced row echelon form, we have the solution for u and v:

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \frac{1}{3} \\ \frac{1}{3} \end{pmatrix} \tag{0.8}$$

Now, we back-substitute to find x and y:

$$u = \frac{1}{x-1} = \frac{1}{3} \implies x - 1 = 3 \implies x = 4, \tag{0.9}$$

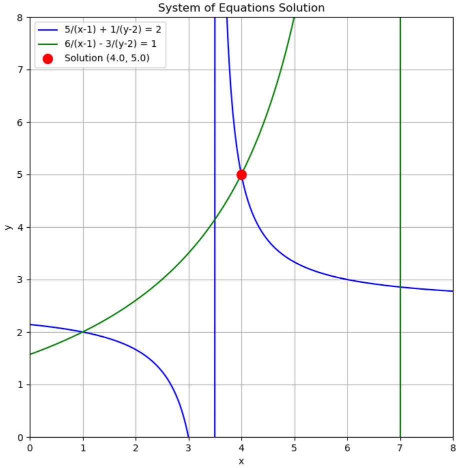
$$v = \frac{1}{y-2} = \frac{1}{3} \implies y - 2 = 3 \implies y = 5. \tag{0.10}$$

Solution: Final Answer

Thus, the final solution is:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}. \tag{0.11}$$

Plot


System of Equations Solution
5/(x-1) + 1/(y-2) = 2
6/(x-1) - 3/(y-2) = 1
Solution (4.0, 5.0)
x
y
0
1
2
3
4
5
6
7
8